

E-Polarized Plane-Wave Scattering by a Finite Sinusoidal PEC Grating via the Nyström Method

A. Petryshyn, S. Dukhopelnykov, and T. Zinenko
O. Ya. Usikov Institute for Radiophysics and Electronics of the
National Academy of Sciences of Ukraine
Kharkiv 61085, Ukraine

Abstract—This paper presents a Nyström formulation for two-dimensional E-polarized plane-wave scattering by a finite sinusoidal grating made of a perfect electric conductor. The inverse-square-root endpoint singularity of the induced current is extracted as a factor, and the logarithmic integral equation is differentiated before the resulting Cauchy-singular equation is discretized at Gauss–Chebyshev nodes. The formulation accommodates arbitrary incidence angles. Convergence under order refinement and agreement with independently implemented analytical-regularization and pulse-basis methods are demonstrated for a flat strip, whose total scattering cross section approaches the geometrical-optics limit at high frequency. Results for a symmetric five-period grating show how corrugation height and electrical size redistribute the absolute differential scattering cross section. Above the first propagating-harmonic threshold, broadened finite-grating lobes appear near the reference directions of the corresponding infinite grating, while finite-length effects preserve a central reflection lobe and an end-generated background. The study therefore clarifies how infinite-grating directions can be used to interpret scattering by short gratings.

Index Terms—Electromagnetic scattering, finite gratings, integral equations, Nyström method, perfect electric conductors.

I. INTRODUCTION

Finite corrugated conductors combine periodic-surface physics with end effects that are absent from an infinite grating. Even a scalar two-dimensional model therefore has to resolve a singular edge current, interaction across several corrugations, and radiation into both halfspaces. Early numerical treatments of two-dimensional diffraction and the associated quadrature techniques are surveyed by Nazarchuk [1]. Nyström algorithms tailored to open strips have subsequently demonstrated controlled convergence for both perfect electric conductor (PEC) and material-strip models [2].

The finite sinusoidal grating has also been studied with complementary methods. Kobayashi and Eizawa used a Wiener–Hopf perturbation analysis for a finite PEC sinusoid [3]; a later H-polarized analysis is reported in [4]. Vinogradova, Kobayashi, and Eizawa later developed a full-wave E-polarized solution by analytical regularization and examined the transition between finite- and infinite-grating behavior [5]. Related resonant configurations formed by two finite sinusoidal gratings are analyzed in [6]. These studies emphasize that periodic Floquet angles remain useful references, but a grating with only a few periods does not possess exact Floquet channels.

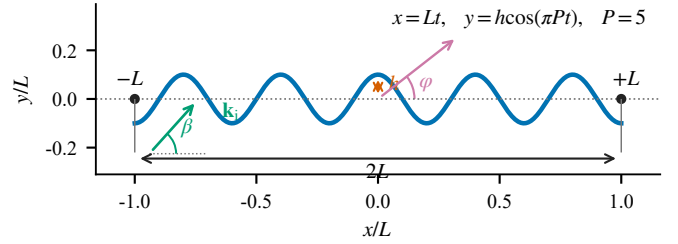


Fig. 1. Symmetric finite sinusoidal PEC grating and angle convention. The incident propagation direction is $\hat{\mathbf{d}} = (\cos \beta, \sin \beta)$; observation angle φ is measured counterclockwise from $+x$.

This paper gives a compact Nyström solution for one finite sinusoidal PEC grating. Its contributions are threefold. First, the geometry is symmetric, all size parameters are referred to the half-span L , and the incident angle is left arbitrary in the formulation. Second, accuracy is assessed by an L^2 density-convergence study and a three-backend comparison. Third, absolute polar patterns and the dimensionally correct total scattering cross section (TSCS) reveal separately the effects of corrugation height and electrical size for a five-period grating.

II. FORMULATION AND NYSTRÖM DISCRETIZATION

A. Geometry and Boundary-Value Problem

For a structure invariant along z , E polarization is represented by the scalar electric component $U = E_z$ in the x – y plane. The zero-thickness PEC arc is parameterized symmetrically as

$$\mathbf{r}(t) = (Lt, h \cos(\pi Pt)), \quad -1 \leq t \leq 1, \quad (1)$$

where $2L$ is its projected width, h/L is the relative corrugation height, and the integer P is the number of complete periods. Its projected period is $\Lambda = 2L/P$. Hence geometry is specified by h/L and P , while frequency enters only through the electrical size kL . The notation and angle convention are summarized in Fig. 1.

With the suppressed time factor $e^{-i\omega t}$, a unit-amplitude incident plane wave is

$$U^{\text{inc}}(\mathbf{r}) = \exp(ik\hat{\mathbf{d}} \cdot \mathbf{r}), \quad \hat{\mathbf{d}} = (\cos \beta, \sin \beta). \quad (2)$$

Thus constant-phase fronts travel along $\hat{\mathbf{d}}$. The reported computations use normal incidence, $\beta = \pi/2$, so the wave travels

from the lower to the upper halfspace; no such restriction is made in the equations. The scattered field U^{sc} satisfies the Helmholtz equation, the outgoing Sommerfeld condition, local energy finiteness at both endpoints, and

$$U^{\text{inc}} + U^{\text{sc}} = 0 \quad \text{on } \Gamma. \quad (3)$$

The outgoing Green kernel is consequently $H_0^{(1)}$.

B. Edge-Conditioned Integral Equation

Absorb the constant in the electromagnetic Green function into a scaled line density q . Its known endpoint behavior is factored as

$$q(t)|\mathbf{r}'(t)| = \frac{v(t)}{\pi\sqrt{1-t^2}}, \quad (4)$$

where v is smooth. Equation (4) extracts the singular factor; it does not remove the physical edge singularity. The scattered field and the logarithmically singular integral equation (IE) are then written using $\rho_t = |\mathbf{r} - \mathbf{r}(t)|$ and $R(t, \tau) = |\mathbf{r}(t) - \mathbf{r}(\tau)|$, with $G(s) = H_0^{(1)}(ks)$, $w(t) = (1-t^2)^{-1/2}$, and $u_0(\tau) = U^{\text{inc}}(\mathbf{r}(\tau))$:

$$U^{\text{sc}}(\mathbf{r}) = \frac{1}{\pi} \int_{-1}^1 G(\rho_t) v(t) w(t) dt, \quad (5)$$

$$\frac{1}{\pi} \int_{-1}^1 G(R(t, \tau)) v(t) w(t) dt = -u_0(\tau). \quad (6)$$

The explicit $1/\pi$ makes the continuous and discrete normalizations transparent.

First, the logarithmically singular IE (6) is differentiated with respect to τ , giving a Cauchy-singular IE. Second, that equation is discretized. In particular, let

$$c_H = -\frac{2i}{\pi}, \quad (7)$$

$$K(t, \tau) = \frac{\partial}{\partial \tau} H_0^{(1)}(kR(t, \tau)) - \frac{c_H}{t - \tau}, \quad (8)$$

$$f(\tau) = -\frac{d}{d\tau} U^{\text{inc}}(\mathbf{r}(\tau)) = -ik U^{\text{inc}}(\mathbf{r}(\tau)) \hat{\mathbf{d}} \cdot \mathbf{r}'(\tau). \quad (9)$$

The coefficient c_H follows directly from the small-argument expansion of $H_0^{(1)}$; with it, K has a finite diagonal limit. The differentiated equation is

$$\frac{1}{\pi} \text{p.v.} \int_{-1}^1 \left[\frac{c_H}{t - \tau} + K(t, \tau) \right] \frac{v(t)}{\sqrt{1-t^2}} dt = f(\tau). \quad (10)$$

Use the Gauss–Chebyshev nodes and the interlaced second-kind points

$$\begin{aligned} t_i &= \cos \frac{(2i-1)\pi}{2N}, & i &= 1, \dots, N, \\ \tau_j &= \cos \frac{j\pi}{N}, & j &= 1, \dots, N-1. \end{aligned} \quad (11)$$

Writing $v_i = v(t_i)$, the implementation returns these samples as `v_nodes`. Thus its publication-facing density already includes the factor π relative to a convention that retains the

quadrature weight π/N . For $j = 1, \dots, N-1$, the boundary rows are

$$\frac{1}{N} \sum_{i=1}^N \left[\frac{c_H}{t_i - \tau_j} + K(t_i, \tau_j) \right] v_i = f(\tau_j). \quad (12)$$

The $1/N$ factor is precisely the Gauss–Chebyshev weight π/N combined with the $1/\pi$ in (10).

Differentiation loses one constant. Equivalence to (6) is restored with one supplementary row,

$$\frac{1}{N} \sum_{i=1}^N M(t_i) v_i = c, \quad (13)$$

where

$$M(t) = \int_{-1}^1 H_0^{(1)}(kR(t, \tau)) \frac{d\tau}{\sqrt{1-\tau^2}}, \quad (14)$$

$$c = - \int_{-1}^1 U^{\text{inc}}(\mathbf{r}(\tau)) \frac{d\tau}{\sqrt{1-\tau^2}}. \quad (15)$$

The logarithmic self term in M is subtracted analytically and its smooth remainder is evaluated by a higher-order Chebyshev rule. Equations (12) and (13) form an $N \times N$ dense system.

C. Far Field and TSCS

For $\hat{\mathbf{s}}(\varphi) = (\cos \varphi, \sin \varphi)$, define Φ_{sc} by $U^{\text{sc}}(\mathbf{r}, \varphi) \sim e^{ikr} \Phi_{\text{sc}}(\varphi)/\sqrt{kr}$. The same quadrature gives

$$\Phi_{\text{sc}}^{(N)}(\varphi) = \sqrt{\frac{2}{\pi i}} \frac{1}{N} \sum_{i=1}^N e^{-ik\hat{\mathbf{s}}(\varphi) \cdot \mathbf{r}(t_i)} v_i. \quad (16)$$

For unit incident amplitude, the differential and total scattering cross sections are

$$\frac{d\sigma}{d\varphi} = \frac{|\Phi_{\text{sc}}(\varphi)|^2}{k}, \quad \sigma = \frac{1}{k} \int_0^{2\pi} |\Phi_{\text{sc}}(\varphi)|^2 d\varphi. \quad (17)$$

Both have the required length dimension. The polar plots below display the absolute quantity $10 \log_{10}[(d\sigma/d\varphi)/L]$; they are not normalized to the maximum of each curve.

For reference, the infinite periodic continuation has tangential harmonics

$$\xi_m = \cos \beta + \frac{m\pi P}{kL}, \quad |\xi_m| < 1. \quad (18)$$

Their upper- and lower-halfspace angles are $\arccos \xi_m$ and $2\pi - \arccos \xi_m$, respectively. These angles are only markers for the broad lobes of the finite grating; below they are called infinite-grating reference directions. At normal incidence this reduces to

$$\cos \varphi_m = \frac{m\pi P}{kL}, \quad \left| \frac{m\pi P}{kL} \right| < 1. \quad (19)$$

The first-order markers become propagating at

$$kL = \pi P = 15.70796 \quad (P = 5). \quad (20)$$

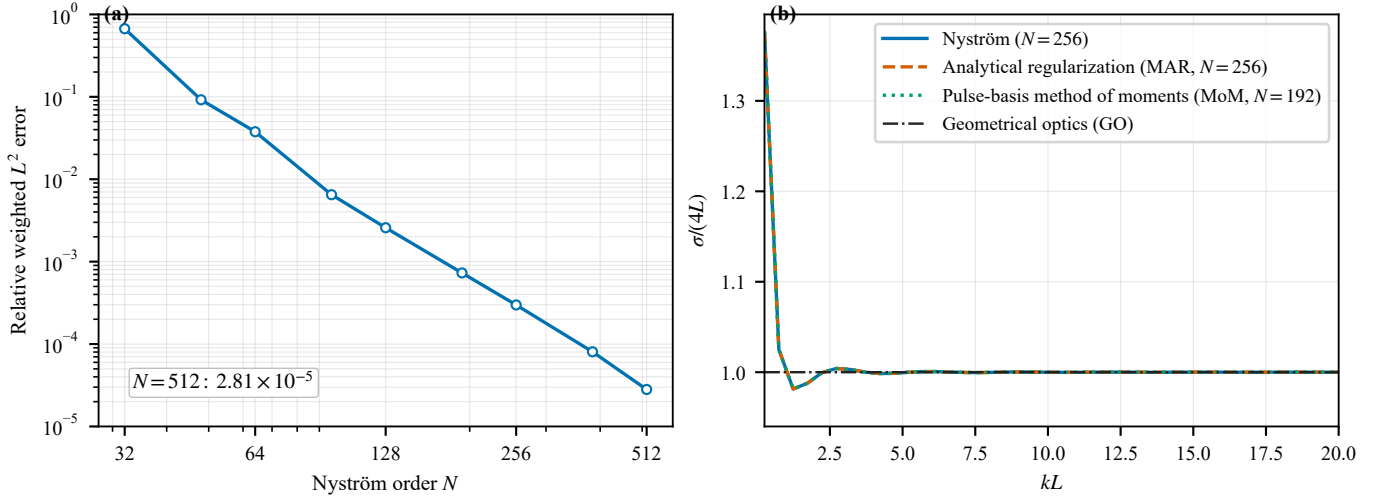


Fig. 2. Verification. (a) Relative L^2 error of the smooth density versus interpolation order for $P = 5$, $h/L = 0.10$, and $kL = 20$. (b) Flat-strip total scattering cross section (TSCS), normalized by its geometrical-optics limit $4L$, from the differentiated Nyström method ($N = 256$), the method of analytical regularization (MAR; $N = 256$), and a pulse-basis method of moments (MoM; 192 elements); the dash-dotted line is the geometrical-optics (GO) limit.

III. NUMERICAL RESULTS

A. Convergence and Cross-Validation

Fig. 2(a) reports a relative weighted L^2 error of the smooth density for $P = 5$, $h/L = 0.10$, $kL = 20$, and $\beta = \pi/2$. Let $I_N v_N$ denote its Chebyshev interpolant and set $\theta_\ell = (\ell + 1/2)\pi/4096$. With the $N_{\text{ref}} = 800$ solution as the reference, the plotted error is

$$\epsilon_N = \left[\frac{\sum_{\ell=0}^{4095} |I_N v_N(\cos \theta_\ell) - I_{800} v_{800}(\cos \theta_\ell)|^2}{\sum_{\ell=0}^{4095} |I_{800} v_{800}(\cos \theta_\ell)|^2} \right]^{1/2}. \quad (21)$$

This midpoint sampling in θ approximates the $L^2_{1/\sqrt{1-t^2}}$ norm in t . For $N = 32, 48, 64, 96, 128, 192, 256, 384, 512$, the error decreases systematically and reaches 2.81142×10^{-5} at $N = 512$. The latter order is used for all corrugated-grating results below. Repeating every unique publication case at $N = 800$ changes the TSCS by at most 7.53315×10^{-6} and the full complex far-field pattern by at most 2.30664×10^{-5} in relative L^2 norm.

The second test uses a flat strip ($h = 0$) over 41 electrical sizes $0.25 \leq kL \leq 20$. Fig. 2(b) compares the present differentiated Nyström implementation with separately assembled analytical-regularization and pulse-basis method-of-moments backends. The maximum spread in $\sigma/(4L)$ among the three results is 1.58902×10^{-5} . The first two use $N = 256$, and the pulse discretization uses 192 elements. At the most demanding endpoint, $kL = 20$, doubling those orders changes the TSCS by at most 5.03819×10^{-6} . At $kL = 20$ the Nyström value is 0.9999612. This is close to the geometrical-optics limit $\sigma/(4L) \rightarrow 1$ for a broadside-illuminated flat PEC strip of projected width $2L$ [2], but it is correctly retained as a finite-frequency value.

B. Representative Five-Period Grating

Fig. 3 shows the normal-incidence solution for $P = 5$, $h/L = 0.10$, and $kL = 20$. The total-field intensity is shown relative to the unit incident-field intensity. Below the grating, incident and reflected fields interfere; above it, the pattern contains the shadow lobe. The polar panel gives the absolute differential cross section over $0 \leq \varphi < 2\pi$. At this electrical size the $m = \pm 1$ periodic-reference harmonics are propagating, and the calculated finite-grating pattern contains broad off-normal structure near their infinite-grating reference directions. The central reflection lobe and end-generated back-ground remain present because only five periods are retained.

C. Height and Electrical-Size Dependence

At fixed $P = 5$ and $kL = 20$, increasing h/L changes the phase and magnitude of the induced current across each corrugation. Fig. 4 shows the resulting redistribution among the specular and off-normal lobes. Because every panel has the same absolute radial scale, changes in level can be compared directly. The normalized TSCS values for $h/L = 0, 0.05$, and 0.10 are, respectively,

$$\frac{\sigma}{4L} = 0.9999612, \quad 0.9948396, \quad 0.9926829. \quad (22)$$

Thus corrugation changes both the angular distribution and the integrated scattered power; the polar curves are not merely rescaled copies.

Frequency changes both the overall electrical width and the set of propagating periodic-reference harmonics. For $P = 5$ and $h/L = 0.10$, the three cases in Fig. 5 have

$$\begin{aligned} \frac{\sigma}{4L} &= 1.010974, & kL &= 12, \\ &= 0.9968797, & kL &= 16, \\ &= 0.9926829, & kL &= 20. \end{aligned} \quad (23)$$

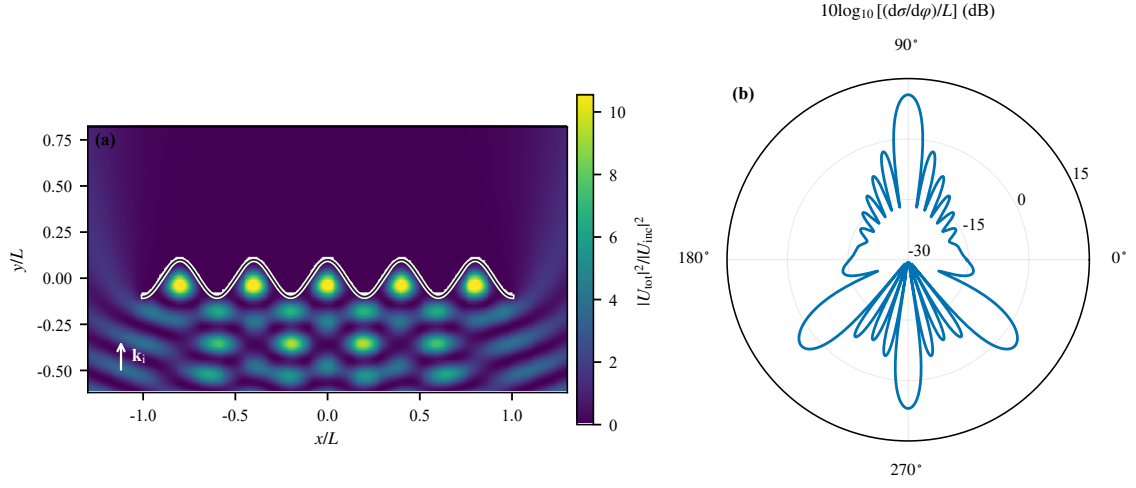


Fig. 3. Five-period grating at $P = 5$, $h/L = 0.10$, $kL = 20$, and $\beta = \pi/2$. (a) Relative total-field intensity $|U^{\text{inc}} + U^{\text{sc}}|^2/|U^{\text{inc}}|^2$. (b) Absolute polar differential scattering cross section, shown as $10 \log_{10}[(d\sigma/d\varphi)/L]$ on the common -30 to $+15$ dB scale.

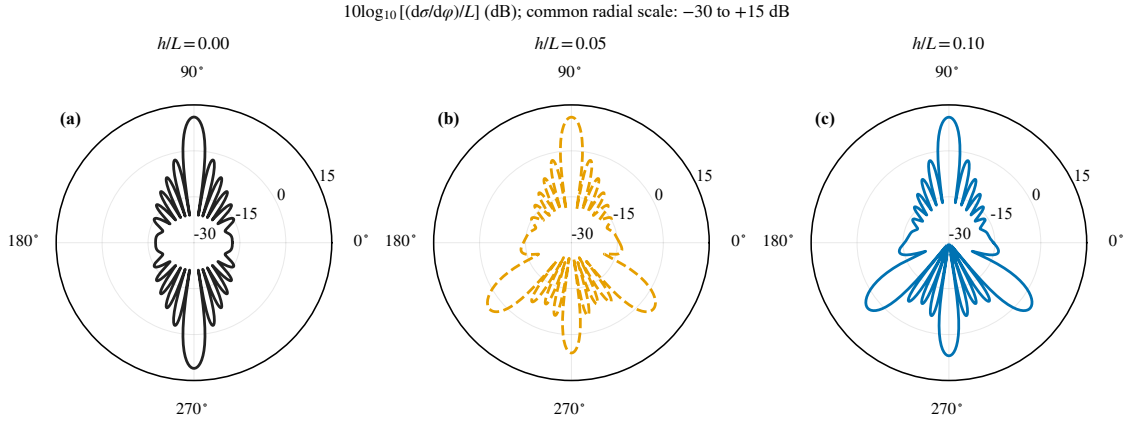


Fig. 4. Absolute polar differential scattering cross section, shown as $10 \log_{10}[(d\sigma/d\varphi)/L]$, for $P = 5$, $kL = 20$, and $\beta = \pi/2$: (a) $h/L = 0$; (b) $h/L = 0.05$; and (c) $h/L = 0.10$. Each panel contains one curve and uses the common -30 to $+15$ dB scale.

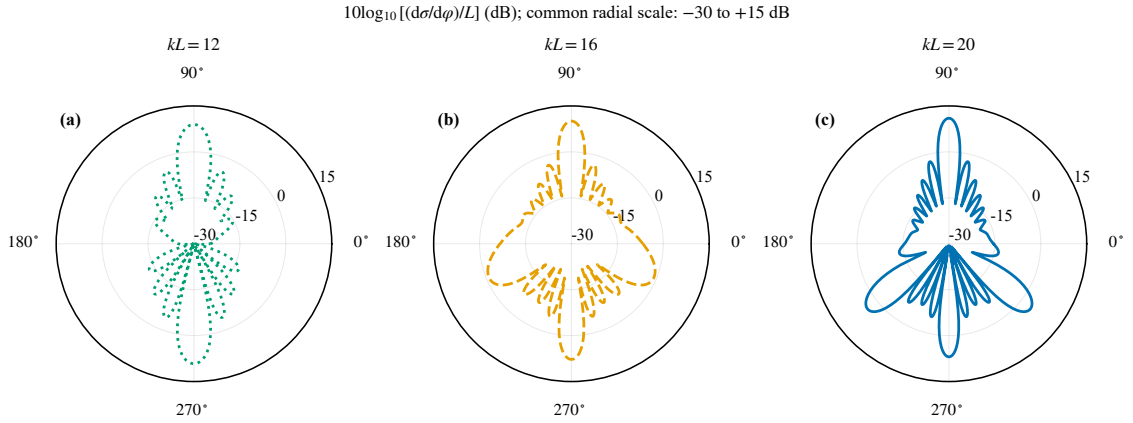


Fig. 5. Absolute polar differential scattering cross section, shown as $10 \log_{10}[(d\sigma/d\varphi)/L]$, for $P = 5$, $h/L = 0.10$, and $\beta = \pi/2$: (a) $kL = 12$; (b) $kL = 16$; and (c) $kL = 20$. Each panel contains one curve and uses the common -30 to $+15$ dB scale. The first-order periodic-reference harmonics are cut off at $kL = 5\pi$ and propagate above it.

At $kL = 12$ the first-order periodic harmonics are evanescent. They become propagating above the threshold in (20), so first-order finite-grating lobes appear in the $kL = 16$ pattern and separate further from their cutoff directions at $kL = 20$. Their finite width and interaction with the background preclude interpreting them as exact diffraction orders.

IV. CONCLUSION

An edge-conditioned integral equation, discretized with the Nyström technique, has been formulated for E-polarized plane-wave scattering by a finite sinusoidal PEC grating. The formulation accommodates arbitrary incidence, uses the outgoing $H_0^{(1)}$ kernel consistently with $e^{-i\omega t}$, and retains the correct $1/k$ factor in the TSCS. Density convergence, agreement among three separately implemented formulations, and the approach of the flat-strip result to its geometrical-optics limit provide complementary accuracy checks. For the five-period grating, height mainly redistributes scattering among angular lobes, whereas electrical size determines when periodic-reference harmonics become propagating. Absolute polar plots and the reported $\sigma/(4L)$ values show that both redistribution and total scattered power vary with the profile.

REFERENCES

- [1] Z. T. Nazarchuk, *Numerical Investigation of Wave Diffraction from Cylindrical Structures*. Kyiv, Ukraine: Naukova Dumka, 1989, in Russian.
- [2] O. V. Shapoval, R. Sauleau, and A. I. Nosich, "Scattering and absorption of waves by flat material strips analyzed using generalized boundary conditions and Nyström-type algorithm," *IEEE Trans. Antennas Propag.*, vol. 59, no. 9, pp. 3339–3346, Sep. 2011, doi: 10.1109/TAP.2011.2161547.
- [3] K. Kobayashi and T. Eizawa, "Plane wave diffraction by a finite sinusoidal grating," *IEICE Trans. Electron.*, vol. E74-C, no. 9, pp. 2815–2826, Sep. 1991, doi: 10.1587/e74-c_9_2815.
- [4] T. Eizawa and K. Kobayashi, "Wiener–Hopf analysis of the H-polarized plane wave diffraction by a finite sinusoidal grating," *Prog. Electromagn. Res.*, vol. 149, pp. 1–13, 2014, doi: 10.2528/PIER14063007.
- [5] E. D. Vinogradova, K. Kobayashi, and T. Eizawa, "Full wave analysis of plane wave diffraction by a finite sinusoidal grating: E-polarization case," *Wave Motion*, vol. 86, pp. 44–62, Mar. 2019, doi: 10.1016/j.wavemoti.2018.12.006.
- [6] E. D. Vinogradova and K. Kobayashi, "Complex eigenvalues of natural TM-oscillations in an open resonator formed by two sinusoidally corrugated metallic strips," *IET Microw., Antennas Propag.*, vol. 15, no. 10, pp. 1283–1298, 2021, doi: 10.1049/mia2.12166.